PRODUCT MANAGER TAKE-HOME · NOMAD VENTURE STUDIO / JOBESCAPE

Answers

Every task in the brief's order, with the calculations.

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Summary — answers to all eleven tasks

Prepared for Nomad Venture Studio (Jobescape). Submitted to @islam_s10.

Answers at a glance

Eleven tasks, in your order. The section number is where the full answer sits in this document.

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The thesis that ties all four parts together

Jobescape has built an excellent machine for selling a product it hasn't finished building.

The funnel acquires an older, serious, surprisingly loyal professional audience at scale. Then **50.2% of paying subscribers never complete a single lesson.** Every part of this assignment lands on the same seam — acquisition is strong, *activation* was never built, and the economics depend entirely on the retention tail that activation would protect.

- **A** — the funnel sells "work faster / feel more confident" to a 45+ professional, and that buyer is loyal (9.5% two-week churn). The product gives them no first win.
- **B** — the content itself is free everywhere. The only defensible moat is scaffolding, feedback and accountability — i.e. precisely the activation layer that is missing.
- **C** — the one feature built to be that layer reaches **11% of payers**, has no daily gate, and moves its target metric by **+0.5 pts (p = 0.82)**.
- **D — 84% of LTV lands after the intro payment** (68% recurring + 16% upsells), so a 1–2 point retention gain beats any funnel squeeze — and refunds/chargebacks are the tail risk.

Part A — Audience & Product

A1 · Segments → part-a-audience/01-segments.md

I split on **driving emotion + job-to-be-done** rather than demographics, because that is what the funnel selects for and what predicts whether a habit product will satisfy the buyer. Six segments — **The Striver · The Latecomer · The Adept · The Optimiser · The Founder · The Switcher** — each tied to a quiz signal, with a full decode of the 37-page quiz funnel.

Derived from real data, twice over. k-modes clustering over 38,071 quiz respondents (from a 49,528-respondent, 210 MB export) and all 9,956 paying subscribers (BigQuery). Three findings that overturned my priors:

- **The audience is old, not young — ~60% of buyers are 45+**, and the largest single cell is men aged 55+ (21.3% of all buyers).
- **Churn falls monotonically with age**: 31.1% (18–24) → 9.5% (45–54). The oldest buyers are the best buyers.
- **The goal a buyer arrives with predicts whether they stay.** "Work faster" and "feel more confident" are half the book and churn at ~11%. "Quick side hustle" churns at **37.2%**. Every creative that promises income buys a customer who leaves.

A2 · The gap → part-a-audience/02-product-gap.md

The underlying gap is a **motivation switch** — the funnel sells fear relief and a fast payoff, the product delivers slow habit-building. Public reviews relocate the *acute* risk to the **price-and-exit gap**: billing above what was agreed, and a refund guarantee gated on finishing the course in ~31 days. The event log confirms the consequence: **11.3% of all unsubscribes are payment failures, chargebacks or refund disputes.** That is regulatory exposure, not ordinary churn — the model the FTC sued Genesis over.

One correction I'd make to the review-sourced picture, having paid and walked the product myself: the widely-repeated "bot-only cancellation" claim is wrong for cancellation. There is a self-serve Manage Subscription tab, and the event log shows 1,358 uses of it. The trap is the refund, not the cancel. What I did find first-hand is a different mismatch: the paywall's "personal AI mentor" is a bot with a stock photo and a human name ("Mark"), and its "24/7 support chat" is a metered 5-credit AI chat.

Part B — Competitors

B1 → 01-competitors.md · **B2** → 02-analysis.md · **B3** → 03-recommendations.md

Jobescape has **structural twins** (Coursiv, Iro AI — "Duolingo for AI") and a cloned funnel (Headway). The decisive fact: **the content is free** — ChatGPT, YouTube, newsletters — so Jobescape cannot win on content. It must win on **scaffolding, feedback and accountability**. Prioritized steals, ordered *de-risk* → *retain* → *differentiate* → *diversify* → *expand*:

1. **Mimo's "Blinkist paywall"** (+100% trial opt-in, −25% early cancels) — cuts refund risk, near-zero build.
2. **Coursiv's branded finite challenge + honest pricing** — the retention container, and it defuses the FTC-style risk.
3. **Iro's Prompt Lab + Section's saveable co-pilot** — the practice surface; the actual moat against free tools.
4. **Owned newsletter + "expensable" pricing** — CAC hedge.
5. **Offer-ladder tests** — activation and ARPU.

Part C — Release Verdict (the Challenge)

C1 · Analytics → 01-analytics.md — 9,956 paying subscribers, all calculations reproducible via `analysis/`.

Reach	89% of payers never start it; 67.5% never see a challenge surface at all
Engagement	of 1,111 starters: 45% complete zero lessons, 14% reach lesson 3, 4.3% finish
Target metric	takers **60.0%** D1 vs people who looked and walked away **59.6%** — **+0.5 pts, p = 0.82**
Unsubscribe	−2.3 pts vs the matched control, p = 0.12 — not significant
CSAT	falls as engagement rises: 3.74 → 3.53 → 3.33 → **3.10** for finishers

C2 · Verdict → 02-verdict.md: **No — the release was not a success.** It moved neither its target metric nor its secondary ones.

Two things make this more than a scorecard:

- **The dashboard would have called it a win.** On the team's own metric definition the Challenge posts 39.8% D1 against a 26.0% baseline — a "+14-point win" that is entirely an artifact of two different clocks and two different populations. Correcting for survivorship takes the gap from +38 pts to +17; matching on exposure takes it to +0.5.
- **v1 never tested its own hypothesis.** The theory was "one day = one skill builds a daily habit," but the gate is a dismissible warning, not a lock, and it is not logged. **27% of the people who finished the "7-day" challenge finished it in a single sitting**, 46% within two. You cannot fail a daily-habit test with a feature that has no days in it.

C3 · What's next → 03-whats-next.md: rebuild the mechanic, then re-run it as a real experiment. Sequenced **instrument** → **mechanic** → **first-ten-minutes** → **quality** → **reach**, with pre-registered success criteria and a kill rule.

The counter-intuitive call: **reach is the last fix, not the first.** The obvious response to "only 11% see it" is to promote it — but satisfaction *falls* as engagement rises, so scaling reach on 3.25-star content scales refunds, which Part D says is the real threat.

C4 · Prototype — ▶ LIVE: https://alyasska.github.io/Nomad_Venture_Studio_TA_C4/

A working React app built in **Jobescape's own design system**, so it reads as a feature of the product rather than a mockup of one. Five steps around **the daily gate — the mechanic v1 never shipped** — each with its own URL and each annotated on the page with the finding that produced it. It takes Jobescape's visual language but not its streak: the buyer is a professional in their fifties, and the 18–24s who respond to streaks and confetti are 9% of the book and churn at 31%. Source: Alyasska/Nomad_Venture_Studio_TA_C4.

Part D — Subscription Economics

D1 · LTV → 01-ltv-model.md

PLAN	MIX	GROSS LTV	NET LTV
1-WEEK	10%	$68.66	**$60.42**
4-WEEK	70%	$140.57	**$123.70**
12-WEEK	20%	$184.26	**$162.15**
Blended	100%	—	**$125.06**

Full model and code in `model/ltv_model.py`. Sensitivity on the horizon question keeps blended net LTV in a tight **$120–$125** band.

Validated against the real cohort: the observed plan mix is 10.2% / 64.6% / 25.1% against the brief's 10 / 70 / 20 — reweighting moves blended net LTV to **$126.90**, so the headline is safe. The 1-week plan is the weak point: lowest LTV, almost no upsell value, and **34.6% churn inside two weeks**.

D2 · A/B break-even → 02-ab-test-model.md — plan-upgrade break-even at **≈4.9%** take-rate. **Recommend running it**; only the $3.60 second-upsell is at risk. Watch upgraded-cohort churn and refunds in the live test.

The three questions I'd ask before doing any of this

Asking the right questions is part of the brief, and these are the ones the data cannot answer:

1. **Was the Challenge ever meant to be judged on D1?** There is no hold-out anywhere in the data. If it shipped as a refund-gate or a support-deflection device, it should be judged against refunds and ticket volume — and my verdict changes.
2. **What D1 number would the team have called success?** I can say the release moved nothing; I can't say whether it missed the bar by a little or a lot.
3. **Which LTV horizon does the team model** — the full supplied curve, or 52 calendar weeks? It is the one genuine ambiguity in Part D, worth $5 of blended LTV.

What's in this submission

1 · Answers	this document — every task in your order, with the calculations
2 · Presentation	48 slides, ~30 minutes, all eleven tasks. Citations are clickable in the PDF
3 · Research dossier	the evidence, the market picture, and the economics thesis
4 · Appendix	generated analysis output, all 8 SQL queries, the verification run, and the key scripts
5 · CV	
Prototype	https://alyasska.github.io/Nomad_Venture_Studio_TA_C4/ — Part C Task 4, live

Every figure in Parts C and D can be re-derived in one command: `cd part-c-release-verdict/analysis && python3 05_qa.py`.

Pre-submit checklist

- [x] Prototype published to GitHub Pages — public, verified in a logged-out browser
- [ ] Attach `Jobescape-Research-Dossier.pdf`, `Jobescape-PM-Deck.html` + `Aliaskar_Bekishev_CV.pdf`
- [ ] BQ password / secrets NOT present anywhere in the submission
- [ ] **Request the $15.19 refund from @islam_s10 — LAST step** (it ends product + BigQuery access)

A1 · Audience segments

How these were produced

Derived, not asserted. k-modes clustering (Huang 1998) over **38,071 quiz respondents × 10 categorical answers**, from the 210 MB quiz export (49,528 respondents; 38,071 answered at least 6 of the 10 core questions). Hamming distance, per-attribute modes, unanswered kept as its own level, seeded, 4 restarts per k. Reproduce with `analysis/cluster_quiz.py`.

Why k-modes and not k-means — a mean over nominal levels ("35–44", "Business owner") has no meaning. One-hot + Euclidean silently asserts that every pair of levels is equidistant, which is exactly the assumption being tested.

> **The honest headline: there is no elbow.**
>
> Within-cluster cost per user falls smoothly across k = 2 … 8 — 5.09, 4.79, 4.67, 4.49, 4.40, 4.31, 4.20 — with no knee. **This population is a continuum, not a set of natural kinds.** Six is a resolution chosen and defended, not a count the data forced. Any segmentation of this audience is a decision about granularity, and should be presented as one.

The six segments

SEGMENT	WHO THEY ARE	SHARE	STRONGEST SIGNAL (LIFT VS POPULATION)
The Striver	senses a gap and is closing it deliberately — growth-motivated, 45–54, male-skewed	28.4%	"somewhat ready, could be behind" **1.9×** · growth-motivated 1.6×
The Latecomer	the oldest buyers, hasn't started, watching the field move without them	22.0%	"falling behind as others move faster" **2.2×** · never used Claude 1.9× · 55+ 1.4×
The Adept	already gets real value from AI and wants more leverage — female-skewed	20.9%	"AI already helps me a lot" **2.5×** · confident 2.5× · female 1.7×
The Optimiser	mid-career professional using Claude on real work, wants speed	12.3%	aged 35–44 **2.6×** · work tasks 1.4× · already uses Claude 1.3×
The Founder	learning AI to build something of their own	9.0%	"start my own business" **3.1×** · female 2.2×
The Switcher	between jobs, changing direction, starting from zero	7.4%	"exploring options" **4.4×** · hasn't tried AI 2.3×

Names follow one pattern deliberately — six agent nouns — so they read as a set and stay memorable in a room. Each is a **runnable quiz filter**, not a persona sketch: every segment above can be reconstructed from answers the funnel already collects.

What this replaced, and what changed

The first version of this analysis was a hand-written rule chain (`segment_sizer.py`): six segments authored from a jobs-to-be-done framework, then counted, with unmatched respondents falling through to a default bucket — which is why one segment came out at 38.7%. **The clustering did not validate it.**

	FINDING
Did not survive	"Gen Z / Early Striver" — 18–24 never separates as a cluster. "Burned Time-Poor Doer" — no cluster at any k.
Was missing entirely	**The Adept (20.9%)** — already competent, female-skewed. The largest group the framework failed to imagine, and commercially the most interesting: a retention-and-expansion audience, not a fear-relief one.
Was cut on the wrong variable	the sharpest single signal in the data is `work_status = "exploring options"` at **4.4× lift** — a *status* variable. The old taxonomy cut that group by *age*, so it dissolved.

Limits

- `sub` / `unsub` / `upsell` are **all zero** in this export, so segments cannot be profiled by conversion here. Buyer-side demographics come from `subscribe_events` in BigQuery (n = 9,956) and are unaffected — see `02-product-gap.md` and Part C.
- No elbow means the cluster boundaries are soft. Treat shares as indicative of *mix*, not as hard memberships.
- k-modes is sensitive to initialisation; 4 restarts per k with a fixed seed makes this run reproducible, not globally optimal.

A2 · Does the product meet the need

The *expectation* side is fully evidenced (ad creatives + the decoded quiz). The *reality* side combines **public user evidence** — App Store (4.7/5, ~1.5K), Google Play (4.4), **Trustpilot (4.5/5, ~8,000 reviews)**, plus the closest structural twin Coursiv as a proxy — with **my own walkthrough as a paying customer** (I bought the 4-week plan for $15.19 and screenshotted the whole journey: `materials/walkthrough/observations.md`). Where the two disagree, I say so. Full evidence + citations: `research/deep-05-product-reality.md`. Confidence tags: **[STRONG]** = triangulated across ≥2 independent surfaces.

> **The tell before the details:** ratings are *high* (4.5–4.7) **and** billing complaints are a constant undercurrent. That combination is the fingerprint of an aggressive review-solicitation funnel sitting on a monetization model that generates a steady stream of disputes. The high score is a marketing asset, **not** proof the gap is closed.

The three-lens gap (now evidence-backed)

LENS	WHAT THE ENTRY POINT PROMISES	WHAT USERS SAY THEY **GOT** (PUBLIC REVIEWS)	GAP
What they expected	Cheap entry (~$25 / "trial"); a personalized plan; expert instruction; exclusive AI tools	A bigger bill — a **+$88 "AI support" charge in the fine print**, "$100 for a year when they wanted a 3-month trial," silent renewals, double-charges (one **bank flagged the charge as fraud**); content "you can simply ask ChatGPT," repackaged free tools, an **anonymous team**	**Price + depth/credibility [STRONG]**
Pre-purchase doubts	"Will this work for me? Am I too behind?" — the quiz *raises* these	Doubts often *reinforced*: "training is shallow vs. free alternatives," "everything... is AI-generated"	Doubt-resolution [STRONG]
Promised-not-delivered	The "earn or money-back" guarantee; easy access; (on one surface) a **freelance income** — "land clients in 3 months"	Refund **gated on completing 14 modules / the plan within ~31 days**; outcome-buyers feel misled ("online gurus promising 10k/mo")	**Exit + outcome [STRONG]**

The positioning inconsistency (a gap-multiplier) [MODERATE]

Jobescape can't decide what it is: **jobescape.me** sells "Master the Claude / AI at work" (a *skills* promise), **app.jobescape.me/academy** sells "Launch and Elevate Your Freelancing Career with AI" (an *income* promise), and the **App Store title keeps rotating** (AI chatbot → AI skills → AI for freelancers → automation). Different users arrive expecting different things — a job outcome vs. a prompting course — so the gap is *pre-loaded* before the product even opens. The income framing is the specific source of the "unrealistic / guru" complaints.

Where the main gap is, and the risk for us

The dominant, evidence-backed gap is not "slow content" — it's the price-and-refund gap: users are billed more than they agreed, and then the money-back promise turns out to have

a condition attached. The refund guarantee that lowers signup friction is **weaponized** — it only pays out if you finish the course/Challenge in a short window. The motivation switch I hypothesized (fear-sale → habit product) is real, but the *acute* risk sits one layer down, in billing and refunds.

Why that's the #1 risk (and why it's easy to miss): - It converts dissatisfied users into **refund disputes, chargebacks, and 1-star reviews** — and a chargeback/fraud flag is far more expensive than a churn. - It is **regulatory, not just reputational.** This is exactly the model the **FTC sued Genesis (a peer quiz-funnel studio) over in June 2026**, and CA's auto-renewal law adds state exposure — an asymmetric tail (the reference **$7.5M ARL settlement ≈ ~3× annual net revenue**, Part D). - **The 4.5–4.7 ratings hide it**, so the org may under-weight the very risk that most threatens the fragile recurring economics (Part D).

The through-line to the rest of the assignment: the fix is the same one Part D calls an economic moat and Part B ranks #1–2 — **transparent pricing + a refund guarantee that isn't a completion trap.** It simultaneously closes the reality gap, defuses the regulatory tail, and lifts voluntary retention. (See also Part C: the Challenge currently doubles as the refund gate — it should earn completion, not coerce it.)

Two corrections I'd make to my own review-sourced read [SEEN]

Paying for the product changed two conclusions, and I'd rather flag them than quietly keep the tidier version:

1. **"Cancellation is bot-only" is wrong.** The paywall fine print names a self-serve **Manage Subscription** tab, it exists, and the BigQuery event log records **1,358 uses of the cancel flow** in a two-week cohort. The bot gates the **refund**, not the cancellation. That is the difference between a dark pattern and a bad guarantee, and only the second one is true.
2. **The gap is wider than the reviews say in one specific place.** The paywall sells "**personal AI mentors** and 24/7 support chat." In-product, the mentor is a **single bot with a stock-photo face and a human name ("Mark")**, and the "24/7 chat" is a **metered 5-credit** AI chat. Nobody in the public reviews names this; I only found it by buying it.

Also worth recording: the live paywall runs **dynamic "61% off" pricing** ($6.93 / $15.19 / $25.99 intro, renewing at $38.95 / $38.95 / $66.65) that differs from the plan table supplied for Part D. Part D uses the brief's figures as instructed; this is flagged as an observation, not a correction to the model.

B1 · Find competitors

How I split direct vs. indirect. *Direct* = solves the same job (get better at using AI for work) for the same audience (non-technical professionals) in a substitutable format. *Indirect* = competes for the same attention, budget, or acquisition channel while differing in subject or format. A third category — **free substitutes** — turns out to be the most dangerous of all, because the *content* Jobescape sells is available for free.

COMPETITOR	TYPE	WHY IT'S A COMPETITOR
Coursiv	**Direct** — structural twin	Same quiz → app → subscription funnel, same "learn AI for your job," same intro-price band. The closest rival.
Iro AI	**Direct** — product twin	Literal "Duolingo for AI." Closest *product*, but weak commercially (soft funnel, ~$60/yr, small brand).
Section (SectionAI)	**Direct** — premium incumbent	Same audience (AI for knowledge workers); credibility via Scott Galloway; now pivoting to an AI-coach SaaS.
Outskill	**Direct** — high-ticket funnel	Free 2-day AI workshop → $1–3k cohort; bids in the same Meta ad auctions.
Be10x	**Direct (near-future)**	Funnel-mechanics twin (₹9 tripwire → mastery upsell); India-first today, **US launch Nov 2025.**
The Rundown / Rundown University	**Direct** — media flywheel	Teaches applied AI; acquires via a 2M+ owned newsletter instead of the ad auction.
Duolingo	**Indirect** — the template	Different subject, but *the* engagement engine Jobescape imitates (streaks / daily habit).
Headway	**Indirect** — GTM twin	Same Meta-ads → quiz → paywall → gamified playbook, applied to microlearning/books.
Mimo	**Indirect** — UX exemplar	Gamified coding app; best-in-class onboarding & paywall mechanics worth copying.
Sololearn	**Indirect → near-direct**	Gamified mobile app **now explicitly teaching AI skills** (prompt engineering, ML).
ChatGPT / Claude themselves	**Indirect substitute**	*The* free self-teaching path — learning by doing at zero cost.
YouTube · Coursera · Udemy	**Indirect substitute**	Free/cheap credible content covering the same skills (Coursera had >3.2M GenAI enrolments in 2024).
AI newsletters (Superhuman AI, Mindstream, The Rundown)	**Indirect**	Own "apply AI in 5 minutes" — almost Jobescape's verbatim promise; 1M+ subs each.
Employer L&D	**Indirect substitute + expansion path**	Free-to-user corporate AI training; also Jobescape's future B2B tier.

The one strategic fact that frames all of Part B: the *content* is free (ChatGPT/Claude, YouTube, newsletters give away exactly what Jobescape teaches), and the *funnel* is already cloned (Coursiv, Iro AI, Headway run the identical playbook). **Jobescape cannot win on content or on funnel novelty — only on scaffolding, feedback, and accountability** (the AI tutor as a *practice surface*, the one thing free substitutes structurally can't provide). That conclusion drives Task 2 and Task 3.

Sources for every competitor (URLs) in `research/03-competitors-direct.md`, `research/04-competitors-indirect.md`, `research/deep-01-teardown-twins.md`, `research/deep-02-teardown-premium.md`.

B2 · How they serve their audience

For each, three questions: *which audience & how close to ours · which intents it addresses and how (mechanics/onboarding/product) · what's done well vs. poorly.* Deep teardowns with sources in `research/deep-01-teardown-twins.md` and `research/deep-02-teardown-premium.md`.

Structural twins & ad-funnel operators

Coursiv — the closest twin

- **Audience:** near-identical — non-technical professionals, fear/FOMO framing, paid-ads traffic. **~95% overlap.**
- **How it serves them:** Meta ad → quiz funnel → web-to-app subscription with a low (~$6–20) tripwire → ~$39.99/4wk recurring. Packages its retention arc as a branded, finite **"28-Day AI Challenge"** with a completion certificate — turning vague AI dread into a *closable* loop with a natural D28 renewal moment.
- **Well:** the funnel machine and the finite-challenge container. **Poorly:** trust — **275+ BBB complaints** over undisclosed recurring pricing (it shows different prices to different users). That's the avoidable failure mode.

Iro AI — the product twin

- **Audience:** same "learn AI for your job," but a soft, no-fear funnel and a small brand → **weak commercial threat.**
- **How it serves them:** an explicit "Duolingo for AI" app — gamified paths, an **AI-IQ test**, custom learning paths, and a **"Prompt Lab"** where you write a real prompt against a live model and get scored/coached. Freemium → ~$9.99/mo or ~$59.99/yr.
- **Well:** the Prompt Lab (active practice) is the best idea in the field — it kills the "too passive/basic" complaint. **Poorly:** no urgency engine, thin monetization, low brand trust. *Use it as a product benchmark, not a monetization rival.*

Outskill — the high-ticket funnel

- **Audience:** same fearful professional, but converts them into a **premium** buyer. **Overlaps in the ad auction.**
- **How:** free 2-day live AI workshop → $1–3k cohort upsell; heavy paid acquisition. **Well:** proves the *same* fear-driven buyer will pay $500–3,000 for a live, time-boxed event. **Poorly:** high-touch, hard to scale; live-event dependency.

Be10x — the tripwire CRO exemplar

- **Audience:** India-first (₹ pricing) today; **US launch Nov 2025** makes it a near-future direct rival.
- **How:** a ~$1 (₹9) workshop tripwire → mastery-program upsell. **Well:** the near-zero price *maximizes registrations and manufactures sunk-cost commitment* ("I paid, I'll show up") — lifting activation for low-intent ad traffic far above a free lead magnet. **Poorly:** geo/pricing not yet transferable to US/EU.

Premium & media incumbents

Section (SectionAI) — the loudest strategic signal

- **Audience:** AI for knowledge workers — the *same* audience, premium end. Credibility via Scott Galloway; acquisition via his ~400–500k **owned newsletter**, not the ad auction.
- **How: has pivoted from courses to an AI-coach SaaS** — tiers now sold on "AI coaching" + saveable, reusable use-cases (Basic free → Premium ~$750/yr). **The tell:** even the premium incumbent admits **courses alone don't retain**, so it's bolting on a recurring *tool*. **Poorly:** premium price limits the mass-market fear buyer.

The Rundown / Rundown University — the media flywheel

- **Audience:** broad applied-AI learners via a **2M+ newsletter**. **How:** free daily newsletter (grown ~$2/subscriber via beehiiv Boosts; moved ~50% of budget off Meta) → **Rundown University ~$1,008/yr**, deliberately priced at the **"expense-it" threshold** (~70% of members bill it to employer L&D) and bundled with "$1,600+ of AI tools" so the price reads net-positive. **Well:** owned-audience acquisition (compounding, cheap) + expensable pricing. **Poorly:** newsletter ≠ structured skill-building; retention rests on habit, not outcomes.

Mimo — the onboarding/paywall playbook (indirect, but the exemplar to copy)

- **Audience:** gamified coding learners (adjacent), but its **UX is the transferable asset. How:** ~15-step commitment-escalating onboarding, slider personalization, early paywall (~1:21 into the session) de-risked with a 14-day trial + social proof; streaks/hearts double as a second monetization surface. Ran the **"Blinkist paywall" experiment (trial-timeline screen + pre-expiry reminder) that doubled trial opt-ins (+100%) and cut 24-hour cancellations 25%. Well:** best-in-class trial/paywall mechanics. **Poorly:** subject (code) differs from ours.

The template & the substitutes

- **Duolingo** — proves the streak is the moat (7-day streak = 2.4–3.6× continuation; streak-freeze cut at-risk churn ~21%). The engagement engine Jobescape imitates. **Duolingo Max** (a paid GPT tutor) is its fastest-growing tier — validating that users *already pay a premium for exactly Jobescape's core product*.

> **Reconciling this with Part C (worth stating, because it looks like a contradiction).** Part B says the streak is Duolingo's moat; Part C recommends *removing* gamification from the Challenge. Both are right, because Duolingo's real lesson is **daily pacing**, not the flame. What retains is that today's lesson is available today and tomorrow's is not — Jobescape's Challenge copied the pageantry (a streak counter, a badge, a certificate) and skipped the pacing, which is why its streak displays 0 and 27% of finishers binge the whole thing in a day. And Duolingo's audience is not Jobescape's: ~60% of Jobescape's buyers are 45+, and the under-25s who respond best to streak mechanics are 9% of the book and churn at 31%. **Take the rhythm, leave the confetti.**

- **Free substitutes (the real threat)** — ChatGPT/Claude *are* the free practice ground; YouTube/Coursera give away the content; newsletters own "apply it in 5 min"; Reddit/Discord own free peer help. **Jobescape must sell scaffolding, feedback, and accountability — not content — because the content is free.**

B3 · What to take, prioritised

The order is the argument. It follows the money (Part D): Jobescape's value is a **fragile recurring tail**, front-loaded and churn-exposed, so the sequence is **de-risk the fragile economics → build retention → build defensibility → diversify acquisition → expand monetization.** Each item names the source competitor, the expected effect, and *why it sits where it does.*

#	TAKE THIS	FROM	EFFORT	MOVES
1	**"Blinkist paywall"** — trial-timeline screen + pre-expiry reminder	Mimo	Very low	Refunds↓, trial→paid↑
2	**Branded finite "Challenge" as the retention container + honest recurring-price disclosure**	Coursiv	Med	Retention↑, refund/legal risk↓
3	**Practice surface** — "Prompt Lab" graded practice + role-specific co-pilot with saveable use-cases	Iro AI + Section	Med	Differentiation, D3/D7↑
4	**Owned newsletter / daily brief as a CAC hedge + "expensable" annual framing**	The Rundown	Med (slow)	CAC↓, break-even↓
5	**Offer-ladder tests** — ~$1 commitment tripwire + high-ticket live "AI Mastermind"	Be10x + Outskill	Med	Activation↑, ARPU↑

1. Copy Mimo's "Blinkist paywall" verbatim — *first, because it's the highest ROI at the lowest cost*

Add a trial-timeline screen ("here's what happens on day 0 / day 12 / day 14") plus a pre-expiry reminder. Mimo measured **+100% trial opt-ins and −25% 24-hour cancellations**. It's a near-zero build, and it directly attacks the **refund/chargeback risk** that threatens the recurring tail (Part D). Stopping the bleeding comes before everything else.

2. Adopt Coursiv's branded finite "Challenge" as the retention container — with honest pricing

Reframe the product around a finite, named challenge arc with a completion certificate and a clear renewal moment (this *is* the Part C v2). Pair it with **transparent recurring-price disclosure + easy cancel** — Coursiv's 275 BBB complaints are the failure mode to avoid, and the FTC's June-2026 suit against Genesis makes this an existential, not cosmetic, fix. Retention + compliance moat in one move.

3. Build the practice surface — Iro's "Prompt Lab" + Section's saveable co-pilot

Turn the existing chatbot from commoditized content-delivery into **graded practice + a role-specific work co-pilot with saveable, reusable use-cases.** This is the moat against the real threat — *free substitutes* — because feedback on *your* attempt is the one thing ChatGPT/YouTube can't give you against Jobescape. Section's own pivot to an AI-coach SaaS is the proof this is where retention lives.

4. Launch an owned newsletter / in-app daily brief — *fourth, because it compounds slowly*

Build an owned audience (à la The Rundown; grow via beehiiv Boosts ~$2/sub + referral) and frame the annual plan as **expensable** to employer L&D. It reduces 100%-paid-ads dependence and improves Part D break-even the moment blended CAC drops — but it takes months to compound, so start early, harvest later.

5. Test the offer ladder — *last, after retention & differentiation are fixed*

A ~$1 **commitment tripwire** (Be10x) to lift activation and manufacture sunk-cost show-up commitment, and a high-ticket live **"AI Mastermind"** (Outskill) to lift ARPU and give paid ads a second, higher-LTV offer to bid against. These are monetization expansions — valuable, but only worth the complexity once the leaky bucket (1–3) is fixed.

Why not lead with acquisition or price? Because Part D shows retention is a bigger, safer lever than the funnel (a 1–2pt retention gain > a near-impossible conversion gain, with none of the regulatory downside). Fixing what leaks before pouring in more traffic is the entire logic of the order.

C1 · Release analytics

Data: BigQuery `persona-496908.sql_assessment` (`app_events` + `subscribe_events`), pulled 2026-07-27. **Cohort:** all **9,956 paying subscribers** whose first app activity falls in **2026-06-12 → 2026-06-25** (14 days). Every user in `app_events` has a purchase row — the product sits entirely behind the paywall, so "user" = "payer" throughout. **Feature:** `course_id = 338` — *7-Day Claude Challenge*, shipped in 6 languages. It is **99.7% of all challenge starts** in the window (1,111 of 1,114), so "the Challenge" and "338" are interchangeable here. **Queries:** `sql/13_comparison_groups.sql` (the one-row-per-user export) and `sql/10` – `12`. **All calculations:** `analysis/` — `02_main.py`, `03_supplement.py`, `04_loose_ends.py`. Every number below is reproducible by running those three files.

The three findings, up front

1. **Reach — 89% of paying subscribers never start the Challenge.** Two-thirds (67.5%) never see a challenge surface at all.
2. **Effect — the target metric does not move.** Users who took the Challenge retain at **60.0% D1**; users who *looked at the Challenge and walked away* retain at **59.6%**. The difference is **+0.5 pts, p = 0.82.** The headline "+38 pt" gap is selection, not the feature.
3. **Mechanic — the gate is soft.** A warning appears if you start the next lesson early, then lets you through. Nothing enforces it and no event records it, so nothing gates the content to one lesson per day. **27% of finishers completed the whole "7-day" challenge in a single day**, 46% within two. The habit engine the hypothesis depends on was never actually shipped.

A fourth finding is the reason "just show it to more people" is the wrong response: **satisfaction falls as engagement rises** — CSAT goes 3.74 (never took it) → 3.53 (low) → 3.33 (high) → **3.10 (finished it)**.

1 · Where the audience goes — the exposure ladder

Before any retention question: how many payers reach the feature at all?

STEP	USERS	% OF ALL SUBSCRIBERS	CONVERSION FROM PREVIOUS STEP
Paying subscribers in the cohort	9,956	100%	—
... saw the Challenge popup	3,140	31.5%	**31.5%**
... clicked the popup	2,970	29.8%	94.6%
... viewed a Challenge page	2,109	21.2%	71.0%
... joined Challenge 338	1,341	13.5%	63.6%
... **started** Challenge 338	**1,111**	**11.2%**	82.8%
... completed ≥1 challenge lesson	612	6.1%	**55.1%**

STEP	USERS	% OF ALL SUBSCRIBERS	CONVERSION FROM PREVIOUS STEP
… completed ≥3 lessons	158	1.6%	25.8%
… completed the whole challenge	48	0.5%	30.4%
… downloaded/shared the certificate	23	0.2%	47.9%

The single biggest leak is the very first step: 68.5% of payers never see the popup. Once someone *does* see it, the funnel is healthy — 94.6% click it. This is a distribution problem, not a desirability problem.

Split of the whole cohort by how far they got:

	USERS	%
Never reached any challenge surface	6,716	67.5%
Reached a surface, never started	2,129	21.4%
Started the Challenge	1,111	11.2%

2 · Inside the Challenge — where starters die

The Challenge ships **8 lessons**, not 7 (every user flagged `course_completed` has exactly 8 completions — a naming/scope mismatch worth fixing on its own).

REACHED	USERS	% OF THE 1,111 STARTERS	SURVIVED THE PREVIOUS STEP
pressed Start	1,111	100%	—
completed ≥1 lesson	612	55.1%	**55.1%**
completed ≥2	308	27.7%	**50.3%**
completed ≥3	158	14.2%	51.3%
completed ≥4	105	9.5%	66.5%
completed ≥5	80	7.2%	76.2%
completed ≥6	62	5.6%	77.5%
completed ≥7	52	4.7%	83.9%
completed all 8	47	4.2%	90.4%

Two cliffs, both at the front. 45% press Start and do nothing at all; half of those who do lesson 1 never do lesson 2. Past lesson 3 the drop-off almost stops — survival climbs from 51% to 90%. The content is not the problem for people who get in; **getting in is the problem.**

3 · Engagement tiers — definition and calibration

The brief asks me to define high and low myself. I use **challenge lessons completed**, because it is the only effort measure that isn't circular with retention (see §5, Test C).

TIER (BRIEF)	MY NAME	RULE	USERS	% OF SUBSCRIBERS
HIGH	Doers + Finishers	started, completed **≥3** of 8 lessons	**158**	1.6%
LOW	No-shows + Samplers	started, completed **≤2**	**953**	9.6%
DIDN'T TAKE IT	Never offered + Passed on it	never started 338	**8,845**	88.8%

Why the cut is at 3. Three reasons, all checkable: - It is where the D7 curve steps — 18% at ≤2 lessons, 39% at ≥3. - It matches the product's own claim: "one day = one skill", so 3 lessons ≈ a 3-day habit. - **It is not cherry-picked.** Sweeping the threshold across every possible value (`02_main.py` §4):

HIGH = ≥ K LESSONS	N HIGH	D1 HIGH	D1 LOW	D1 GAP	UNSUB HIGH	UNSUB LOW
1	612	61.4%	58.3%	+3.1	12.7%	9.8%
2	308	63.3%	58.8%	+4.5	13.0%	10.8%
3	**158**	**69.6%**	**58.4%**	**+11.2**	11.4%	11.4%
4	105	74.3%	58.5%	+15.7	10.5%	11.5%
5	80	78.8%	58.6%	+20.2	10.0%	11.5%
6	62	80.6%	58.8%	+21.8	11.3%	11.4%
7	52	76.9%	59.2%	+17.7	11.5%	11.4%
8	47	76.6%	59.3%	+17.3	12.8%	11.4%

The retention gap grows monotonically with the cut — and **the unsubscribe gap is ~0 at every threshold.** Whatever separates high from low engagement, it does not show up in the metric that pays the bills.

4 · The headline comparison

All three groups anchored on **each user's first app day**, so they are on the same clock. (§5 Test B explains why this matters and what the team's own anchor does to the numbers.) Brackets are 95% Wilson intervals.

METRIC	HIGH (3+ LESSONS)	LOW (0–2 LESSONS)	DIDN'T TAKE IT
D1 retention ★ target	**69.6%** [62.1–76.3]	**58.4%** [55.3–61.5]	**21.7%** [20.9–22.6]
D3 retention	57.0% [49.2–64.4]	34.4% [31.5–37.5]	10.0% [9.4–10.6]
D7 retention	39.1% [31.2–47.6]	18.0% [15.4–21.0]	6.9% [6.3–7.5]
Unsubscribe rate	11.4% [7.3–17.3]	11.4% [9.6–13.6]	14.3% [13.6–15.1]
(D7 denominator)	*133*	*737*	*6,345*
Avg lessons completed (all courses)	19.28	6.38	1.81
Avg active days in product	5.37	3.46	1.76
Avg CSAT (1–5)	**3.33**	**3.53**	**3.74**

METRIC	HIGH (3+ LESSONS)	LOW (0–2 LESSONS)	DIDN'T TAKE IT
(CSAT raters)	*158*	*845*	*3,740*

Significance vs "didn't take it" (two-proportion z-test):

COMPARISON	DIFF	Z	P	
HIGH — D1	+47.9 pts	+14.27	<1e-15	***
LOW — D1	+36.7 pts	+24.76	<1e-15	***
HIGH — D7	+32.2 pts	+13.92	<1e-15	***
HIGH — unsubscribe	−2.9 pts	−1.04	0.298	ns
LOW — unsubscribe	−2.9 pts	−2.43	0.015	*

Read literally, this table says the Challenge triples D1 retention. It does not. §5 is the actual analysis.

5 · Is the gap causal? — four tests

An "engaged users retain better" result is what selection produces on its own: motivated users both take challenges *and* come back. Four ways to break the claim.

Test A — exposure-matched control *(the decisive one)*

"Didn't take it" is not one group. Most of them were never offered the feature. The fair comparison is people who **reached the Challenge and chose not to engage**.

EXPOSURE GROUP	N	D1	D3	UNSUBSCRIBE
Started 338	1,111	**60.0%** [57.1–62.9]	37.6%	11.4%
Joined 338, never started	230	**60.4%** [54.0–66.5]	37.8%	13.9%
Viewed 338, never joined	764	**59.3%** [55.8–62.7]	36.6%	13.6%
↳ the two combined = the control	**994**	**59.6%**	36.9%	13.7%
Saw the popup, never opened 338	1,132	44.9% [42.0–47.8]	24.0%	**25.5%**
Never reached a challenge surface	6,716	12.2% [11.4–13.0]	3.7%	12.5%

> **Takers vs exposed-but-didn't-start: D1 diff +0.5 pts, z = +0.22, p = 0.823 (ns).** Unsubscribe diff −2.3 pts, p = 0.119 (ns).

People who joined the Challenge and never opened a lesson retain **exactly as well** as people who completed it. The predictive power sits entirely in *reaching the challenge surface* — i.e. in being an engaged user — not in the Challenge doing anything.

Tightened further: holding total product activity constant, the effect stays at zero (`04_loose_ends.py`).

ACTIVE DAYS IN PRODUCT	N TAKERS	D1 TAKERS	N CONTROL	D1 CONTROL	DIFF	P
1	97	0.0%	63	0.0%	+0.0	1.000
2	270	46.3%	257	42.4%	+3.9	0.370
3–4	446	64.8%	427	65.1%	−0.3	0.924
5–7	217	82.9%	205	84.4%	−1.4	0.689
8+	81	90.1%	42	76.2%	+13.9	0.038

Four of five strata are flat. The one significant cell is the smallest (n = 81 vs 42) and is one of five tests — it does not survive multiple comparisons.

Test B — immortal-time bias

Only **24% of takers start the Challenge on their first app day**; the median taker starts on **day +1**, and 28% start on day +3 or later. To start on day *k* you must already have survived to day *k*. Takers are pre-selected survivors — the comparison is rigged before it begins.

GROUP	N	D1	D3
Takers who started on **day 0**	266	**39.1%** [33.4–45.1]	22.6%
All takers (any start day)	1,111	60.0% [57.1–62.9]	37.6%
Never took it	8,845	21.7% [20.9–22.6]	10.0%

Removing the survival head-start alone cuts the apparent gap from **+38 pts to +17 pts**. Test A removes the rest.

Test C — a non-tautological dose-response

"More lessons → better retention" is partly circular: doing lesson 3 on day 3 *is* retention. The clean version: among the 816 takers whose entire challenge activity happened **on one day**, does a bigger day-0 dose predict coming back tomorrow? Nothing circular here.

LESSONS DONE ON DAY 0	N	D1 THE NEXT DAY	UNSUBSCRIBE
0	429	55.7% [51.0–60.3]	10.5%
1	248	54.4% [48.2–60.5]	12.5%
2	99	50.5% [40.8–60.1]	11.1%
3	21	33.3% [17.2–54.6]	23.8%
4	5	0.0%	20.0%
8 (binged the whole thing)	13	53.8% [29.1–76.8]	**38.5%**

> **≥2 vs ≤1 day-0 lessons: D1 diff −9.2 pts, z = −1.98, p = 0.048.**

The dose-response is flat-to-negative. Consuming more Challenge content on day 0 does not make you come back — and the users who consumed *all* of it unsubscribed at 38.5%. Which is the mechanic problem:

Test D — is it even a daily challenge?

The hypothesis was *"one day = one skill builds a habit of returning every day."* That only works if the content is gated to one lesson per day. It isn't.

- **48 users completed the whole challenge.** Days they took to do it: **1 day → 13 (27%)**, 2 days → 9, 3 days → 6, 4 → 4, 5 → 5, 6 → 2, 7 → 3, 8 → 3, 9 → 2, 10 → 1.
- **46% finished the "7-day" challenge in two days or fewer.**
- **Mean lessons per active challenge day: 1.46.**
- Of the 612 starters who completed at least one lesson, only **225 (36.8%) ever came back for a second challenge day.**

There is no daily gate, no scheduled unlock, and no reason to return tomorrow rather than finish tonight. **v1 did not test the hypothesis** — it shipped a short course with "7-Day" in the title.

6 · The target metric as the team defined it

The brief defines D1 as *"retention the day after the challenge started."* That anchor produces a very different number:

MEASURE	N	RATE
Takers · D1 anchored on challenge-start **(team's definition)**	1,111	**39.8%** [36.9–42.7]
Takers · D3 anchored on challenge-start	982	25.9%
Takers · D7 anchored on challenge-start	606	14.7%
Takers · D1 anchored on first app day	1,111	60.0%
Whole cohort · D1, first app day	9,953	**26.0%**

This is how the release gets misread. On the team's own dashboard the Challenge shows **39.8% D1** against a product baseline of **26.0%** — a apparent +14-point win. But those are different clocks *and* different populations. Put on the same clock and matched on exposure, the effect is zero.

7 · Segments

Age — and the one place I'd push back on the team's own read

AGE	SUBSCRIBERS	%	TAKE RATE	D1 (ALL)	UNSUBSCRIBE
55+	3,200	32.1%	10.2%	25.4%	**9.7%**
45–54	2,551	25.6%	10.9%	26.9%	**9.5%**
35–44	1,887	19.0%	12.0%	27.8%	14.5%
25–34	1,117	11.2%	12.4%	24.4%	19.7%
18–24	913	9.2%	10.3%	21.9%	**31.1%**

Unsubscribe falls monotonically with age. The oldest buyers are the best buyers — 45+ churn at roughly a third the rate of under-25s, and they are the majority of the book.

Age × gender (paying subscribers):

AGE	MALE	FEMALE	OTHER/NA	TOTAL	%
18–24	602	310	1	913	9.2%
25–34	613	493	11	1,117	11.2%
35–44	1,040	815	32	1,887	19.0%
45–54	1,444	1,065	42	2,551	25.6%
55+	**2,120**	1,035	45	**3,200**	**32.1%**
Total	**5,948 (59.7%)**	**3,824 (38.4%)**		9,956	

> **On "our core market is 40–50-year-old men":** the gender half is right — **59.7% of buyers are men.** The age half understates it. **58–60% of paying subscribers are 45+**, and the single largest cell in the entire book is **men aged 55+ (2,120 users, 21.3% of all buyers)** — larger than men 35–54 combined (2,484, 24.9%) relative to any other single cell. This matches Part A's independent finding from the quiz export: of the 39,070 respondents who answered the age question (out of 49,528 total), **61.6% are 45+**.
>
> **Why it matters for this release:** a gamified streak-and-badge mechanic is designed for the audience Jobescape *doesn't* have. The buyer is a 55-year-old man who wants to stop feeling behind at work — competence and dignity framing, not confetti. It also matters commercially: the 45+ majority is the low-churn, high-LTV core, and the under-25s who respond best to gamification churn at 31%.

Goal — the ads that sell the worst customers

STATED GOAL	SUBSCRIBERS	%	UNSUBSCRIBE
Work faster	3,015	30.3%	**10.9%**
Feel more confident with AI	1,992	20.0%	**10.7%**
Start my own business	1,161	11.7%	10.9%
Earn more	830	8.3%	10.7%
Get a promotion / better job	636	6.4%	14.0%
Gain flexibility or work remotely	492	4.9%	**27.4%**
Get a quick side hustle	368	3.7%	**37.2%**
Unlock better income opportunities	347	3.5%	**25.4%**

The "get-rich-quick" intents churn at **2.5–3.5×** the core. They are only ~12% of the book, but they are a disproportionate share of refunds and disputes. This is Part A's positioning finding showing up as money.

Plan

PLAN	SUBSCRIBERS	%	UNSUBSCRIBE
4-Week	6,341	63.7%	12.2%
12-Week	2,495	25.1%	10.6%
1-Week	**1,014**	**10.2%**	**34.6%** [31.8–37.6]
4-Week special	90	0.9%	3.3%

The 1-week plan churns at **3× everything else** — inside a 14-day window, i.e. mostly before or at its first renewal.

Useful cross-check for Part D: the real plan mix is **10.2% / 64.6% / 25.1%** against the brief's assumed **10% / 70% / 20%** — close enough to validate the LTV model, with a mild skew toward the 12-week plan that would nudge blended LTV slightly **up**.

Work status

Full-time employee 33.6% (unsub 11.8%) · Business owner 18.3% (9.4%) · Freelancer/self-employed 14.9% (10.5%) · Exploring options 8.5% (18.6%) · Between jobs 5.8% (13.8%) · Student 1.8% (**37.1%**).

Country

US 37.7% · GB 10.2% · AU 6.9% · CA 5.3% · IT 3.8% · AE 2.6% · DE 2.6% · FR 2.5% · SG 2.4% · ES 2.3%. Tier-1 English-speaking is ~60% of the book, consistent with the brief's stated market.

8 · Unsubscribes and satisfaction

14.0% of the cohort unsubscribed inside 14 days (1,393 of 9,956). Because the window is only 14 days, no 4-week subscriber can even reach a renewal decision — **every unsubscribe rate in this document is a floor, not a final value.**

Stated reasons (recorded for 37% of unsubscribes):

REASON	N	% OF UNSUBS
(none recorded)	873	62.7%
Support request	310	22.3%
hard_decline	84	6.0%
account_deletion	31	2.2%
Mastercard Alert	30	2.2%
general	22	1.6%
dispute / paypal_refund / Visa CDRN / fraud / Merchanto / payment_refunded / fraud_reported	43	3.1%

157 of 1,393 unsubscribes (11.3%) are payment failures, chargebacks or refund disputes — 1.6% of the whole paying cohort in two weeks. That is the Part A/Part D risk showing up in the event log.

The satisfaction inversion

GROUP	RATERS	MEAN CSAT	% RATING 4–5	% RATING 1–2
Never took the challenge	3,740	**3.74**	59.8%	17.5%
LOW (0–2 lessons)	845	3.53	53.0%	20.6%
HIGH (3+ lessons)	158	3.33	38.6%	19.6%
Completed the whole challenge	48	**3.10**	37.5%	22.9%

Challenge-338 content specifically rates **3.25** (583 raters); the *same users* rate the rest of the product **3.45**. **The more of this feature someone consumes, the less they like the product.** Any plan that starts with "drive more traffic into the Challenge" would push more people down this curve.

Does completion protect revenue? No.

Part A found (from public reviews) that finishing the course is the money-back gate. If completion were genuinely valuable we'd expect finishers to churn least; if it's a refund gate we'd expect the opposite. The data says **neither — completion is simply uncorrelated with churn**: finishers 12.5% vs non-finishers 11.4% (p = 0.812, ns).

One signal inside it: finishers who **binged in ≤2 days unsubscribe at 22.7%** (n = 22) while those who **spread it over 3+ days unsubscribe at 3.8%** (n = 26). Small numbers, so I hold this loosely — but it is consistent with everything above: *pace* is what matters, and the feature does nothing to control pace.

9 · What I'd flag before anyone acts on this

1. **There is no hold-out group.** Challenge starts occur on every single day of the window; there is no pre/post period and no control arm. Every causal statement here is an observational correction, not an experiment. **This is the #1 process fix, and it belongs *before* the next release, not after.**
2. **The window is 14 days.** D7 is unobservable for 27.5% of the cohort (denominators are stated everywhere). All unsubscribe rates are floors.
3. **Two quiz vocabularies are pooled in one cohort.** `status = "Full-time worker"` (n = 731) churns at **28.5%** while `status = "Full-time employee"` (n = 3,341) churns at **11.8%** — the same concept in different words, from what must be a different funnel or traffic source. Likewise a stray `age = "45+"` bucket (n = 235) alongside `45-54` / `55+`. These cannot be pooled blindly, and someone should find out which funnel variant produces the 28.5% cohort.
4. **The popup carries no challenge identifier,** so "saw the popup" means exposure to any challenge. Since 338 is 99.7% of starts, the practical impact is negligible.
5. **"7-Day Challenge" ships 8 lessons.** Minor, but it means the product's own promise and its content are off by one before anyone even opens it.

→ **Task 2 — the verdict**

C2 · Verdict

The verdict

No. The release was not a success. On its own stated target metric — D1 retention — the Challenge produced **no measurable effect**. Users who took it retain at 60.0% D1; users who reached the same screen and walked away retain at 59.6% (**+0.5 pts, p = 0.82**). Holding product activity constant, the effect stays at zero in four of five strata. It also did not move unsubscribes (−2.3 pts, p = 0.12).

But "not a success" is not the same as "the idea is wrong," and the distinction is the whole decision. v1 never actually tested its own hypothesis:

> *The hypothesis was "one day = one skill builds a habit of returning every day." Nothing in the shipped feature gates content to one lesson per day. 27% of the people who finished the "7-day" challenge finished it in a single day; 46% within two. The mean is 1.46 lessons per active challenge day.*

You cannot fail a daily-habit test with a feature that has no days in it. What shipped was a short course with "7-Day" in the title.

Decision: do not scale it, do not kill it. Rebuild the mechanic and re-run it as a real experiment. The sequencing — and why reach is the *last* fix, not the first — is Task 3.

The number that would have fooled us

This matters more than the verdict itself.

On the team's own definition — *"D1 = retention the day after the challenge started"* — the Challenge posts **39.8% D1** against a product-wide baseline of **26.0%**. That reads as a **+14-point win**, and any dashboard built to the brief's spec would have reported it as one.

It is an artifact of two mistakes stacked on top of each other:

1. **Different clocks.** Takers are measured from the day they started the challenge; everyone else from their first day in the app. Only 24% of takers start on day 0 — the median starts on day +1, and 28% start on day +3 or later. To start on day *k* you must already have survived to day *k*. That survivorship is baked into the metric.
2. **Different populations.** "Everyone else" is 88.8% of the book, two-thirds of whom never even reached a challenge surface. Comparing an opt-in group to a base that includes people who barely opened the app measures who opted in, not what they opted into.

Correct for the first and the gap falls from +38 pts to +17. Correct for the second and it falls to **+0.5 pts**.

The process lesson is bigger than this feature: the release shipped to 100% with no hold-out. There is no control arm and no pre/post period anywhere in the data. Everything above is an observational reconstruction of an experiment that should have been run properly the first time — and it only worked because there happened to be a large "looked and left" group to use as a natural control. Next time there may not be.

The release was scored against a secondary metric

Added after speaking with a Jobescape PM on 2026-07-28 (notes).

The brief names **D1 retention** as the Challenge's target. The team's actual hierarchy is:

	METRICS
Primary	Gross profit · unsub % at **12h / 24h** · rebill rate **0→1** and **1→2** · LTV
Secondary	D1 / D3 / D7 retention · session time · session depth · CSAT

> *"Gross profit is the first priority; retention is the second."* — and on LTV: *"it's very insensitive, it rarely reacts to anything except the obvious."*

So the feature was scoped against a second-order outcome before a line of it was written. That is a resource-allocation finding, not an analytics one, and it makes the verdict firmer rather than softer: the Challenge missed its stated secondary target **and** shows no effect on the nearest primary-metric proxy the data allows — unsubscribe, −2.3 pts, $p = 0.12$.

What this dataset can and cannot see, checked rather than assumed:

PRIMARY METRIC	OBSERVABLE HERE?
Gross profit	**No** — no revenue, cost or refund-amount fields in either table
Unsub % at 12h / 24h	**Not in my pull** — needs the unsubscribe timestamp; the query is written and ready (`sql/14_unsub_timing.sql`)
Rebill 0→1, 1→2	**No** — `pr_webapp_subscription_renewed` fires **17 times** across 9,956 subscribers; a 4-week plan's first rebill lands at day 28, outside the 14-day window
LTV	**Modelled, not observed** — and by the team's own read, too insensitive to detect a single feature
Unsubscribe, any time in window	**Yes** — and it does not move

I would rather say that plainly than quietly substitute D1 for gross profit and let the deck imply I measured the thing that matters.

And on LTV's insensitivity — here is the size of it, computed from the company's own plan table (`ltv_sensitivity.py`): a **one-point** gain in rebill 0→1 moves blended net LTV by **$1.31 (≈1%)**. On a single cohort that is inside the noise — yet at 1,000 signups a day the same point is worth about **$478,000 a year**. The money is large and the signal is faint simultaneously. That is the argument for reading a Challenge v2 on **unsub 12h/24h and rebill 0→1**, and using LTV only to price the result afterwards.

Which metrics I relied on, and how I read them

METRIC	WEIGHT	HOW I READ IT
D1 retention (target)	Primary	Right metric, wrong denominator. Only meaningful against an exposure-matched control — which shows no effect.
Unsubscribe	Primary (economic)	The metric that actually pays. No significant movement at any engagement threshold. Right-censored by the 14-day window, so all values are floors.
D3 / D7	Supporting	Move with engagement, but circularly — completing lesson 3 on day 3 *is* retention. Treated as descriptive, not evidential.
Lesson completion	Diagnostic	The most useful thing in the dataset: it locates the failure (55% of starters never finish lesson 1) rather than just sizing it.
CSAT	Diagnostic, and a red flag	3.25 for challenge content vs 3.45 for the same users elsewhere, and it *falls* as engagement rises: 3.74 → 3.53 → 3.33 → 3.10 (finishers).

The decision rule I set before looking: **a real success is a smooth dose-response that survives an exposure-matched comparison.** It failed both. The exposure-matched gap is zero, and the one non-circular dose-response test — among users whose entire challenge activity happened on one day, does a bigger day-0 dose predict returning tomorrow? — comes back **flat-to-negative** (−9.2 pts, $p = 0.048$).

Why it turned out this way

Four mechanism-level causes, in order of how much they explain.

1. The habit engine was never switched on

The whole theory rests on daily gating: today's skill is available today, tomorrow's tomorrow, and the gap between them is what builds the return habit. The shipped feature has no gate, no scheduled unlock, and no reason to come back tomorrow instead of finishing tonight. So the behaviour it produced is the behaviour you'd predict from an ungated course: people either binge it or abandon it. **Only 36.8% of the starters who completed a lesson ever came back for a second challenge day.**

This is why the honest verdict is "untested," not "disproven."

2. It was aimed at the wrong layer of the funnel

The Challenge was built as a **retention** feature. The product's constraint at that moment is **activation**:

- **50.2% of paying subscribers never complete a single lesson anywhere in the product.**
- Mean active days in the product: **1.98**. Cohort D1: 26.0%. D7: 8.6%.
- **89% of payers never start the Challenge**; 67.5% never see a challenge surface at all.

A feature reaching 11% of payers cannot move a company-level metric. Even granting the entire (non-causal) raw gap, the arithmetic ceiling is **+4.3 points of cohort D1**. The real effect is indistinguishable from zero, so the realised contribution is ~0.

3. The content doesn't earn the second visit

45% of the people who press Start complete zero lessons. That is the moment of highest intent in the entire feature — they clicked a popup, opened the page, joined, and pressed a button — and nearly half get nothing out of it. Then half of those who *do* finish lesson 1 never open lesson 2.

And satisfaction moves the wrong way: the more of this feature someone consumes, the lower they rate the product (3.74 → 3.10). This is the finding that makes "just promote it harder" actively dangerous — **scaling reach on 3.25-star content scales refunds**, and 11.3% of unsubscribes in this cohort are already payment failures, chargebacks or refund disputes.

4. The mechanic is designed for an audience Jobescape doesn't have

Streaks, badges and daily challenges are built for the demographic that responds to gamification. Jobescape's buyer is not that person:

- **59.7% of paying subscribers are men; 58–60% are 45 or older.** The single largest cell in the book is **men aged 55+ (21.3%)**.
- **Unsubscribe falls monotonically with age** — 31.1% (18–24) → 9.5% (45–54) → 9.7% (55+). The older majority is the loyal, high-LTV core; the under-25s who respond best to gamified mechanics churn at three times the rate.

A 55-year-old professional who bought this because they feel behind at work is not chasing a streak. They want to stop feeling behind — competence and dignity, not confetti. The v2 has to carry the same *structure* the Challenge provides while changing what it rewards.

(This is also the one place I'd gently push back on the team's own read of the audience: "40–50-year-old men" is right on gender and understates age by about a decade. Same conclusion, sharper aim.)

What would change my mind

I would revise to "success" on any one of these:

- A **hold-out cohort** showing a D1 lift for the exposed arm — the only evidence that settles this properly, and the reason Task 3 leads with instrumentation.
- A **positive dose-response that survives exposure matching**, i.e. the effect I looked for and did not find.
- Evidence that the Challenge's job was never D1 at all — e.g. it was shipped as a **refund-gate or support-deflection** device, in which case it should be judged against refunds and ticket volume, not retention. (Worth asking: the data neither confirms nor refutes it — completion is simply uncorrelated with churn, p = 0.81.)
- A **longer window.** 14 days cannot see a 4-week subscriber's first renewal. If the Challenge's benefit is a 30-day effect, this dataset structurally cannot detect it.

What this verdict does *not* say

To be fair to the team that shipped it: the concept is sound and the audience needs exactly this kind of structure — a finite, guided, low-ambiguity path is the right shape of product for a 45+ professional who feels behind. The people who *did* reach it converted through the funnel at 94.6% from popup to click, which says the offer is attractive. **The failure is in the**

build and the placement, not the idea. That is why the recommendation is a rebuild rather than a kill — and why the first thing to fix is the ability to know whether the next one worked.

→ **Task 3 — what's next + Task 4 — prototype**

C3 · What's next · C4 · Prototype

Verdict → action. The Challenge produced no measurable D1 effect, but it also never tested its own hypothesis: it has no daily gate, so there was no habit mechanic to evaluate. The concept is right for this audience — a finite, guided, low-ambiguity path is exactly what a 45+ professional who feels behind at work needs. The build and the placement are wrong.

Decision: rebuild the mechanic, then re-run it as a real experiment. Do not scale reach yet.

The reframe

Today the Challenge is a **retention feature** bolted to the side of the product, reached through a popup, taken by 11% of payers.

It should be the product's **activation path**. The evidence:

- **50.2% of paying subscribers never complete a single lesson anywhere.** The product has no on-ramp; people pay and then face an open-ended catalogue.
- The Challenge is already the right *shape* of on-ramp — finite, sequenced, one skill at a time.
- Where it is offered, people want it: **94.6% of everyone who saw the popup clicked it.** Desire is not the constraint.

So the job of v2 is not "make the challenge stickier for the people who take it." It is **"make the challenge the thing that turns a payer into a user."**

The counter-intuitive part: reach is the LAST fix, not the first

The obvious response to "only 11% see it" is "promote it." That would be a mistake, and the data says why:

Satisfaction falls as engagement rises — CSAT 3.74 (never took it) → 3.53 (low) → 3.33 (high) → **3.10 (finished it)**. Challenge content rates **3.25** while the same users rate the rest of the product **3.45**.

Driving 9,000 payers into 3.25-star content would scale complaints, refunds and chargebacks — which are already 11.3% of unsubscribes in this cohort — and Part D shows refund/regulatory exposure is the biggest threat to the economics. **Fix the thing before you point the firehose at it.**

Order of operations: **instrument → first-10-minutes → mechanic → quality → reach.**

> **This order changed after the PM conversation** (notes). I originally put the day gate before the first-ten-minutes fix, because the gate is the untested hypothesis. But the team's primary metrics are **gross profit, unsub % 12h/24h, rebill 0→1 and 1→2, and LTV** — D1 is explicitly secondary. Fixing the first ten minutes attacks **unsub 12h/24h directly**, which is a primary metric and the cheapest item in the plan. The day gate's nearest primary-metric effect — surviving to the day-28 rebill decision — is one step further out. So the cheap primary-metric fix goes first.

Which metric each phase is actually for:

PHASE	MOVES	TIER
0 · Instrument	Makes "which initiatives win or lose" answerable at all	*enables everything*
1 · First ten minutes	**Unsub % 12h / 24h** → gross profit	**primary**
2 · Mechanic (the day gate)	Survival to the day-28 decision → **rebill 0→1**	**primary**, one step out
3 · Quality	CSAT, session depth	secondary
4 · Reach	Multiplies whatever phases 1–3 achieve	*multiplier*

The team already ships features to **specific cohorts**, so a hold-out is not new machinery — it is how they already work. That removes the only real objection to Phase 0.

Phase 0 — Make the next release measurable *(before any feature work)*

The single most expensive thing about v1 is not that it didn't work; it's that **we can't prove whether it did.** 100% rollout, no hold-out, no pre/post. Everything in Task 1 is an observational reconstruction that only worked because a large "looked and left" group happened to exist as a natural control.

#	CHANGE	WHY
0.1	**10–20% hold-out arm** on the next release	The only design that answers the question. Non-negotiable.
0.2	**Redefine the target metric**: exposure-matched D1 on a common clock	The current definition reports 39.8% vs a 26.0% baseline — a +14pt "win" that is entirely an artifact.
0.3	Log `challenge_day_unlocked` / `challenge_day_missed`	Without them the daily gate is unmeasurable, which is how v1 got here.
0.4	Extend the analysis window past the first renewal	A 14-day window structurally cannot see a 4-week subscriber's renewal decision.

Phase 1 — Fix the first ten minutes

45% of the people who press Start complete zero lessons. That is 499 users in two weeks, at the single highest-intent moment in the feature — they clicked a popup, opened a page, joined, and pressed a button. Nearly half then got nothing.

#	CHANGE	MOVES	EFFORT
1.1	**Find out what happens on that screen first.** Is it a load failure, an auth wall, an app-download step, a long intro? This is a bug hunt before it is a design task.	activation	XS
1.2	**Guaranteed win inside 10 minutes** on a **pre-loaded** example — the blank page is the enemy for this audience	D1, CSAT	M
1.3	**Bridge lesson 1 → 2** (a 50% drop): end lesson 1 with tomorrow's task already loaded	D1	S

1.1 is the highest expected value in the entire plan, costs an afternoon, and lands on a **primary** metric.

Phase 2 — Build the mechanic that was missing

#	CHANGE	MOVES	EFFORT
2.1	**Gate the content to one day at a time.** Tomorrow's skill unlocks tomorrow.	**D1**, D3	S
2.2	**Wire the reminder channel that already exists to the Challenge** — anchored to a user-chosen time, with escalating D1/D3/D7 copy	**D1**	XS
2.3	**Name tomorrow as a concrete work task,** not a topic — "Tomorrow: turn a 40-page PDF into a one-page brief"	D1	XS
2.4	**Catch-up, not punishment** — a missed day can be recovered the next day without losing progress	D3, unsub	S

2.1 is the whole experiment. Everything else in this document is optional next to it: the hypothesis is that daily pacing builds a return habit, and the shipped feature paced nothing.

> **Most of Phase 2 is already built — it just isn't connected.** Walking the paid product myself (`walkthrough/observations.md`), the Automation tab already ships an **active WhatsApp agent**, **"Motivational messages"** and **"Study progress reminders"**, and both Challenges and Academy already render a **streak counter and a Mon–Sun weekly tracker**. The streak displays **0**. So this is not "add a habit system" — it is *"connect the habit system you already shipped to the feature that needs it."* That is why 1.2 is XS effort rather than S, and it makes Phase 2 a wiring job more than a build.

The obvious objection — "gating reduces consumption" — is answered by the data: finishers who binged in ≤2 days unsubscribe at **22.7%**; finishers who spread it over 3+ days unsubscribe at **3.8%** (n = 22 / 26, so directional, not conclusive). Bingeing does not look like the behaviour worth protecting. Guardrail: measure total lesson completions in the test arm and stop if it collapses.

Phase 3 — Fix what people are actually rating 3.25

#	CHANGE	MOVES	EFFORT
3.1	**Mine the exit-reason and bug events** — `lesson_exit_reason_click` (3,601 events), `lesson_bug_click` (1,295), `lesson_bug_text_submitted` (475). A free qualitative dataset on exactly why people quit, already in the warehouse, unanalysed.	CSAT	XS
3.2	**Re-aim the framing at the real buyer.** 59.7% male, 58–60% aged 45+, largest single cell men 55+. Competence and dignity, not streaks and confetti: "you will never fear a spreadsheet again," not "🔥 3-day streak!"	CSAT, D3	S
3.3	**Make the AI tutor the practice surface** — type your real work task, get a usable draft, save it. This is the one thing free ChatGPT doesn't give this user: scaffolding and a worked example for *their* task.	CSAT, D7	M

Phase 4 — Only now, reach

#	CHANGE	GATE
4.1	Make the Challenge the **default post-onboarding path** for new subscribers rather than a popup	Ship only once challenge CSAT ≥ 3.45 (the product baseline)
4.2	Retire the popup as the primary entry point	After 4.1 proves out in the hold-out test

What to remove

- **The "7-Day" name on an 8-lesson course.** The promise and the payload are off by one before anyone opens it.
- **The certificate** — 23 downloads across the entire cohort (0.2%). It is not the reward this audience wants, and it costs build and screen space.
- **More gamification.** The audience that responds to it (18–24) churns at 31.1% and is 9% of the book.

Pre-registered success criteria for v2

Stated before the test so the result can't be reinterpreted afterwards:

	METRIC	BAR
Primary	**Unsub % at 12h / 24h**, test vs hold-out	the metric the business runs on
Primary	**Rebill 0 → 1**, test vs hold-out	worth ~$1.31 of blended LTV per point
Secondary	Exposure-matched D1 lift	**≥ +5 pts**
Secondary	Starters completing ≥1 lesson	55% → **75%**
Secondary	Starters reaching ≥3 lessons	14% → **30%**
Guardrail	Challenge CSAT	**≥ 3.45**, must not fall
Guardrail	Refund / chargeback rate	must not rise
Guardrail	Total lessons completed per starter	must not fall (gating risk)

And the kill rule: if the daily gate ships with a proper hold-out and still shows no lift, kill the Challenge. The conclusion would then be that this audience does not want a challenge, and the activation problem needs a different shape entirely. Naming that in advance is the point — v1's real failure was being unfalsifiable.

Task 4 — the prototype

▶ **Live link:** *(published below — see `prototype/`)*

A mobile-web prototype of the v2 flow, built to make the **riskiest hypothesis testable before eng commits a sprint**: that daily pacing plus a guaranteed early win converts a payer into a returning user.

Five screens, each tied to a finding in Task 1:

SCREEN	THE FINDING IT ANSWERS
1 · Commit — pick your daily time, echo back the goal from the quiz	1.2 — there is currently no scheduled return trigger at all
2 · Today — one unlocked day, tomorrow visibly **locked** with a countdown	1.1 — the missing daily gate; 27% of finishers binged the whole "7-day" challenge in a day
3 · The win — pre-loaded real work task → AI draft → save to your library	2.2 / 3.3 — 45% of starters complete zero lessons; blank page is the enemy
4 · Done for today — result you can use, named next task, gate closes	2.3 — half of lesson-1 finishers never open lesson 2
5 · Tomorrow — the return trigger firing, and the locked day opening	1.3 — a concrete reason to come back rather than a badge

How it improves the next release. It turns the v2 from a document into something you can put in front of five 50-year-old professionals on a Tuesday and watch. Specifically it de-risks three decisions that would otherwise be argued in a room: whether the **daily gate reads as helpful or as being locked out of something you paid for** (the one real risk in Phase 1); whether the **pre-loaded first task** produces a genuine "oh, that's useful" inside ten minutes; and whether **competence framing beats streak framing** for a 45+ audience. Each is a copy/UX question that costs a day to test on the prototype and a sprint to discover in production.

The AI responses are simulated — a sandboxed page can't call a model API. The source marks exactly where a live call drops in. See `prototype/README.md`.

D1 · Total LTV

Answer up front (primary model):

PLAN	MIX	GROSS LTV	NET LTV
1-WEEK	10%	$68.66	**$60.42**
4-WEEK	70%	$140.57	**$123.70**
12-WEEK	20%	$184.26	**$162.15**
Blended (mix-weighted)	100%	—	**$125.06**

Model: `model/ltv_model.py` (reads `data/plans.csv`). All figures reproducible by running it.

How the model works

Per plan I model a cohort of **1 subscriber** and sum the cash they generate over the horizon, then apply the fee.

1. Survival. `Cmn` = probability a user who survived to period *m* pays for period *n*. Everyone pays the intro (period 0, survival `S0 = 1`). Survival to recurring period *k* is the running product:

```
S_k = C01 · C12 · … · C_(k-1,k)
```

2. Revenue (gross).

```
Gross = Intro Price
      + Recurring Price · Σ S_k        (k = 1…12, the 12 renewal transitions given)
      + E[first upsell]  = conv₁ · price₁
      + E[second upsell] = conv₂ · price₂
```

Upsells are one-time at checkout, so their expected value is applied once to every subscriber.

3. Net. Payment-provider fee is **12% on all collected cash** (intro, recurring, upsells):

```
Net = Gross · (1 − 0.12)
```

Worked example — 4-WEEK plan (the 70% plan)

- Survival: S_1=.67, S_2=.4355, S_3=.3049, S_4=.2286, then ×0.80 each → S_{12}=.0384. **Σ S_k = 2.400.**
- Recurring revenue = 39.99 × 2.400 = **$95.98**
- Intro = **$19.99**
- Upsells = 0.30×69.99 + 0.12×29.99 = 20.997 + 3.599 = **$24.60**
- Gross = 19.99 + 95.98 + 24.60 = **$140.57** → Net = ×0.88 = **$123.70**

Assumptions (stated explicitly — the brief is silent on these)

1. **Fee base.** 12% applies to *all* revenue including upsells (they clear the same processor). If the fee were subscription-only, net LTV rises slightly.
2. **Everyone pays the intro** (S0 = 1); the C-curve begins at the intro→first-recurring step (`C01`).
3. **Horizon = the 12 supplied transitions.** The data gives exactly 12 renewal steps per plan. For the 4-week plan that is 13 periods × 4 weeks = **exactly 52 weeks**, which anchors the "one-year horizon." I apply the full 12-step curve to every plan as the **primary** model.
4. **Upsells fire once, at checkout**, independent of retention.

Sensitivity — strict 52-calendar-week horizon

The one wrinkle: a **12-week** plan billed 12 times spans ~2.8 years, not one. Under a strict "only payments within 52 weeks" reading, the 12-week plan gets just **4** billing periods:

PLAN	NET LTV (PRIMARY, FULL-CURVE)	NET LTV (STRICT 52-WK)
1-WEEK	$60.42	$60.42
4-WEEK	$123.70	$123.70
12-WEEK	$162.15	**$135.53**
Blended	**$125.06**	**$119.74**

Only the 12-week plan (20% of mix) moves — under the strict reading it gets 4 charges (weeks 12/24/36/48, all inside the year) instead of 12 — so blended net LTV sits in a tight **$120–$125** band either way. I lead with $125.06 but flag the band. **Which reading the reviewer intends is the one open question I'd confirm** (see submission notes).

Validation against the real cohort (BigQuery)

The brief supplies the plan table as given, so the model uses it as instructed. But the Part C pull lets me check whether those inputs describe reality — **9,956 paying subscribers**, 2026-06-12 → 2026-06-25 (`part-c-release-verdict/analysis/04_loose_ends.py`).

Plan mix — the model's weights hold up:

PLAN	`PLANS.CSV` ASSUMES	OBSERVED
1-WEEK	10%	**10.2%**
4-WEEK	70%	**64.6%** *(incl. a 0.9% `4Week_special` variant)*
12-WEEK	20%	**25.1%**

The mild shift from 4-week to 12-week moves blended net LTV **up**, not down: reweighting to the observed mix gives **$126.9** against the modelled $125.06. The headline number is safe.

Early churn — directionally consistent, and the 1-week plan is the weak point. Observed unsubscribe inside the 14-day window: **1-week 34.6%, 4-week 12.2%, 12-week 10.6%.** Only the 1-week plan can even reach a renewal decision inside that window, and its 34.6% sits in the right neighbourhood of the 45% first-period drop the supplied `C01 = 55%` implies (the gap is users who lapse passively rather than pressing cancel). The 4-week and 12-week figures are floors — their first renewals fall outside the window entirely — so this validates the shape of the curve, not its level.

What it changes: nothing in the answer, but it sharpens the read-through. The 1-week plan is 10% of subscribers, generates the lowest net LTV ($60.42), carries almost no upsell value ($0.72/sub), and churns at 3× the others. It is the plan most exposed to the refund/chargeback risk in the economics thesis — worth asking whether it earns its place in the lineup at all.

Read-through

- The **4-week plan carries the business** (70% of subs, $123.70 net each).
- The **12-week plan is the most valuable per user** ($162 net) but only 20% of mix — a natural target for a mix-shift, which is exactly what Task 2 tests.
- **Upsells matter unevenly:** $24.60/subscriber on the 4-/12-week plans vs just $0.72 on the 1-week plan (its upsell prices are tiny — $1.99/$0.99).

D2 · A/B plan-upgrade break-even

Question: In the test group, a **4-week** subscriber is shown a **Plan Upgrade to 12-week** ($49.99 one-time) **instead of the second upsell**. An upgrade buyer is billed under 12-week economics from then on. At what upgrade-conversion does the test beat control, and should we run it?

Answer up front: Break-even upgrade conversion is ≈ **4.9%** (primary model) to ≈ **8.3%** (conservative horizon). Both are low and realistic for a checkout upgrade offer → **yes, run the test.**

What changes between control and test (4-week buyers only)

COMPONENT	CONTROL	TEST
4-week intro ($19.99)	✅	✅ (unchanged)
First upsell (30% × $69.99)	✅	✅ (unchanged)
Second upsell (12% × $29.99 = $3.60)	✅	❌ **replaced**
Plan Upgrade ($49.99, one-time)	—	✅ taken by fraction *p*
Recurring stream	4-week ($39.99 curve)	*p* switch to **12-week** ($62.99 curve); (1−*p*) stay 4-week

Intro and first upsell are identical in both arms, so they cancel in the comparison. Because the 12% fee is uniform, it also cancels — **break-even is identical gross or net.**

The break-even formula

Setting test LTV = control LTV and cancelling the common terms:

```
 p* · (UpgradePrice + RecRev₁₂ − RecRev₄) = E[2nd upsell]₄
p* = 3.60 / (49.99 + RecRev₁₂ − RecRev₄)
```

Each upgrade **adds** the $49.99 surcharge plus the *difference* between a 12-week and a 4-week recurring stream, and **cannibalises** only the $3.60 expected second upsell.

Primary model (full 12-period curve)

- Recurring revenue, 4-week stream = **$95.98**
- Recurring revenue, 12-week stream = **$119.67** (Δ = **+$23.69**)
- Each upgrade adds 49.99 + 23.69 = **$73.68**, cannibalises **$3.60**
- **Break-even p*** = 3.60 / 73.68 = **4.88%**

UPGRADE TAKE-RATE	TEST NET LTV (PER 4-WK BUYER)	LIFT VS CONTROL ($123.70)
0% (control)	$123.70	—
5%	$123.77	+0.1%

UPGRADE TAKE-RATE	TEST NET LTV (PER 4-WK BUYER)	LIFT VS CONTROL ($123.70)
10%	$127.02	+2.7%
20%	$133.50	+7.9%
30%	$139.98	+13.2%

Conservative model (strict 52-week horizon)

If the 12-week recurring stream is capped at payments charged inside one calendar year (4 periods — weeks 12/24/36/48 — $89.43), an upgrade slightly *shortens* the modelled recurring window, so the surcharge does more of the work:

- Each upgrade adds 49.99 + (89.43 − 95.98) = **$43.44**, cannibalises $3.60
- **Break-even p*** = 3.60 / 43.44 = **8.28%**

Even here the test turns positive by ~8% take-rate and reaches +6.7% lift at 30%.

Recommendation

Run the test. The downside is tiny and bounded: the only thing at risk is the $3.60 expected second upsell per 4-week buyer, and only for the test arm. The upside is a permanent **mix shift toward the highest-LTV plan** ($162 vs $124 net). Break-even sits at **~5–8%** upgrade conversion — well within what checkout upgrade/one-click-upsell offers typically achieve.

How I'd run it & de-risk: 1. Size the test for the target metric = **upgrade take-rate**, with net-LTV-per-4-week-buyer as the readout. Need enough 4-week buyers to detect a ~5% take-rate with confidence. 2. **Watch second-order effects the point model ignores:** (a) does adding an upgrade step depress overall checkout completion / first-upsell take? (b) do upgraded users **churn faster** because they were pushed into a bigger commitment (higher refund/chargeback risk)? Track early-period retention and refunds on upgraded cohorts, not just the day-0 take-rate. 3. Because break-even is so low, the real question isn't "will it pay off at a given conversion" but "does the upgrade **cannibalise something more valuable** (checkout completion, refunds)" — that's what the live test, not the spreadsheet, has to answer.

Model & scenarios: `model/ltv_model.py`.